



The Genetic Architecture of Wisdom: *A Genome-Wide Association Study in 131,870 Individuals*

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The Genetic Architecture of Wisdom Remains Unmapped

SD-WISE-7 Scale: We used the SD-WISE-7 scale [1], developed by Thomas, Jeste, and colleagues at UC San Diego. It captures seven core components:

- 1. Emotional Regulation:** I remain calm under pressure.
- 2. Self-reflection:** I avoid self-reflection. (R)
- 3. Ability to accept divergent perspectives:** I enjoy being exposed to diverse viewpoints.
- 4. Decisiveness:** I tend to postpone making major decisions as long as I can. (R)
- 5. Social Advising:** I often don't know what to tell people when they come to me for advice. (R)
- 6. Spirituality:** My spiritual belief gives me inner strength.
- 7. Prosocial behaviors:** I avoid situations where I know my help will be needed. (R)

The Gap: Higher SD-WISE-7 scores correlate with health and well-being, but the trait's genetic architecture remains entirely unmapped.

Cohort Demographics:

Phenotype	Mean $\pm$ SD, range or %	Pearson r	P-value	N
Age (mean, SD)	52.82 (15.92)	0.143	<1E-308	142398
Age (range)	18-90			
Female	120395 (70.66%)	-0.052	3.13E-87	142398
Education (mean, SD)	15.01 $\pm$ 2.43	0.209	<1E-308	141719
Income (annual, mean, SD)	129,151.03 $\pm$ 99,259.02	0.158	<1E-308	134460
Married	94497 (55.46%)	0.059	8.53E-107	136370

Wisdom is Heritable and Polygenic

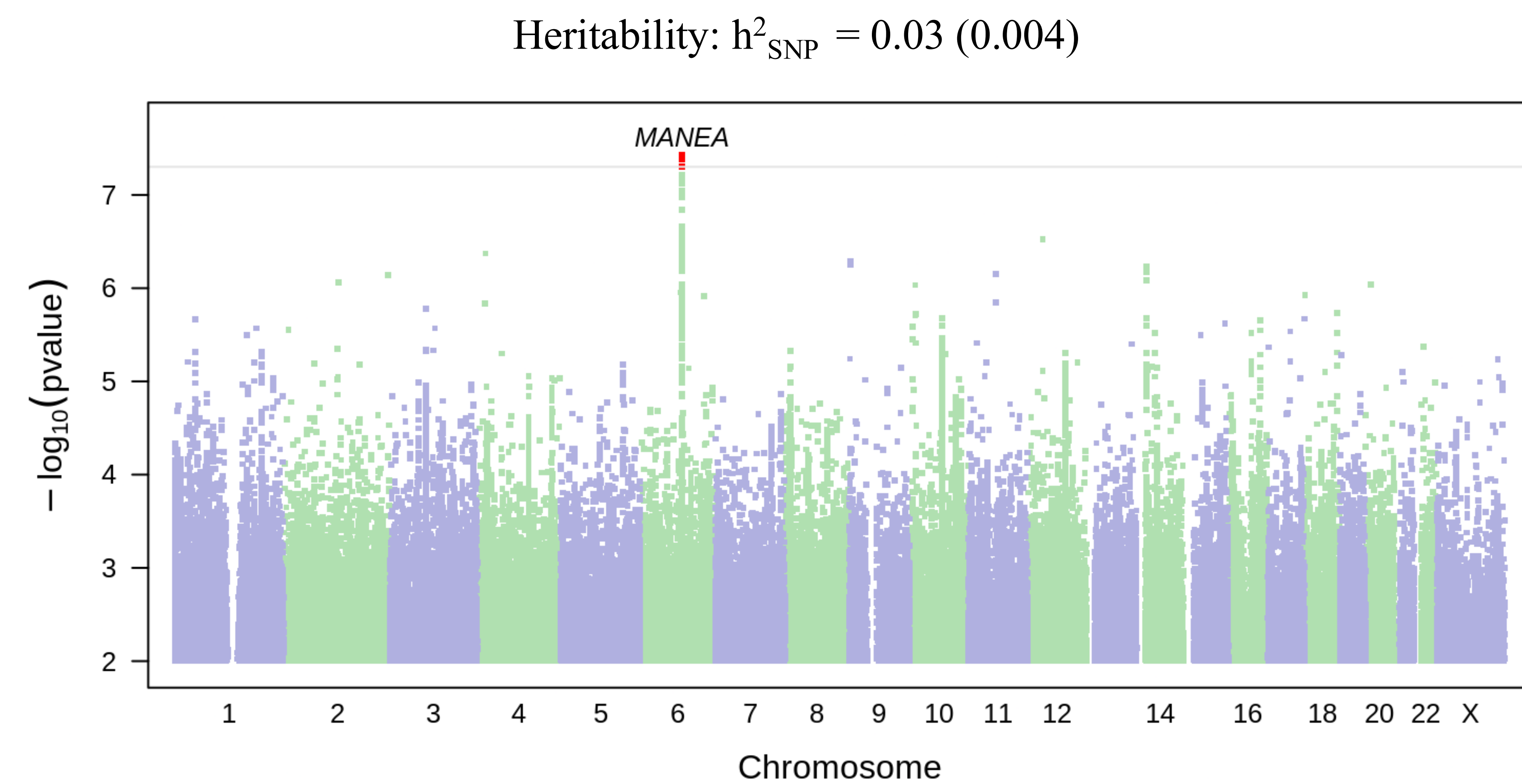


Figure 1: Mapping the Genetics of Wisdom. A genome-wide association scan of 131,870 individuals identifies the first significant genetic locus for wisdom on chromosome 6 (Rs1028640, $P=3.7\text{e-}08$, in the *MANEA* gene).

Wisdom is Genetically Correlated with Character and Well-Being, Not Cognition

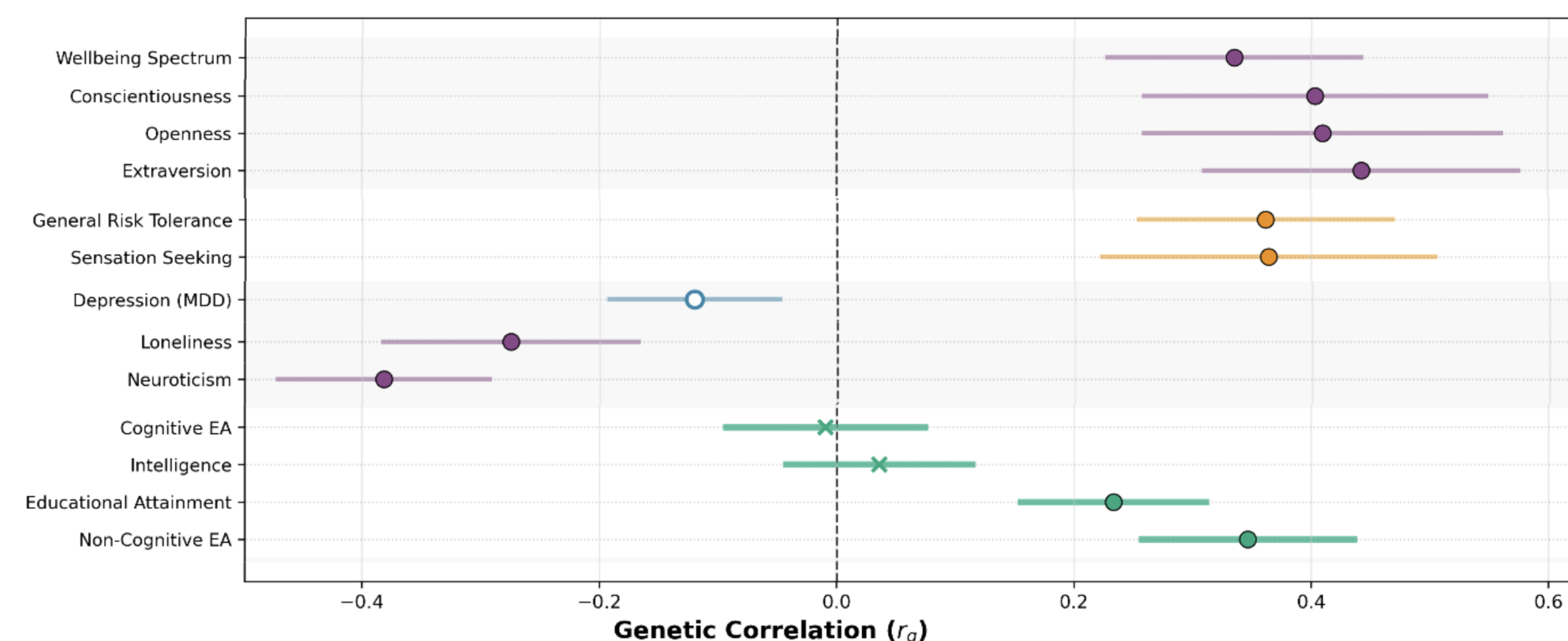


Figure 2: Genetic Overlap of Wisdom. Wisdom shares significant genetic architecture with emotional stability and positive character traits but is entirely distinct from cognitive capacity.

Wisdom polygenic scores are associated with a range of health and well-being outcomes in *AoU*

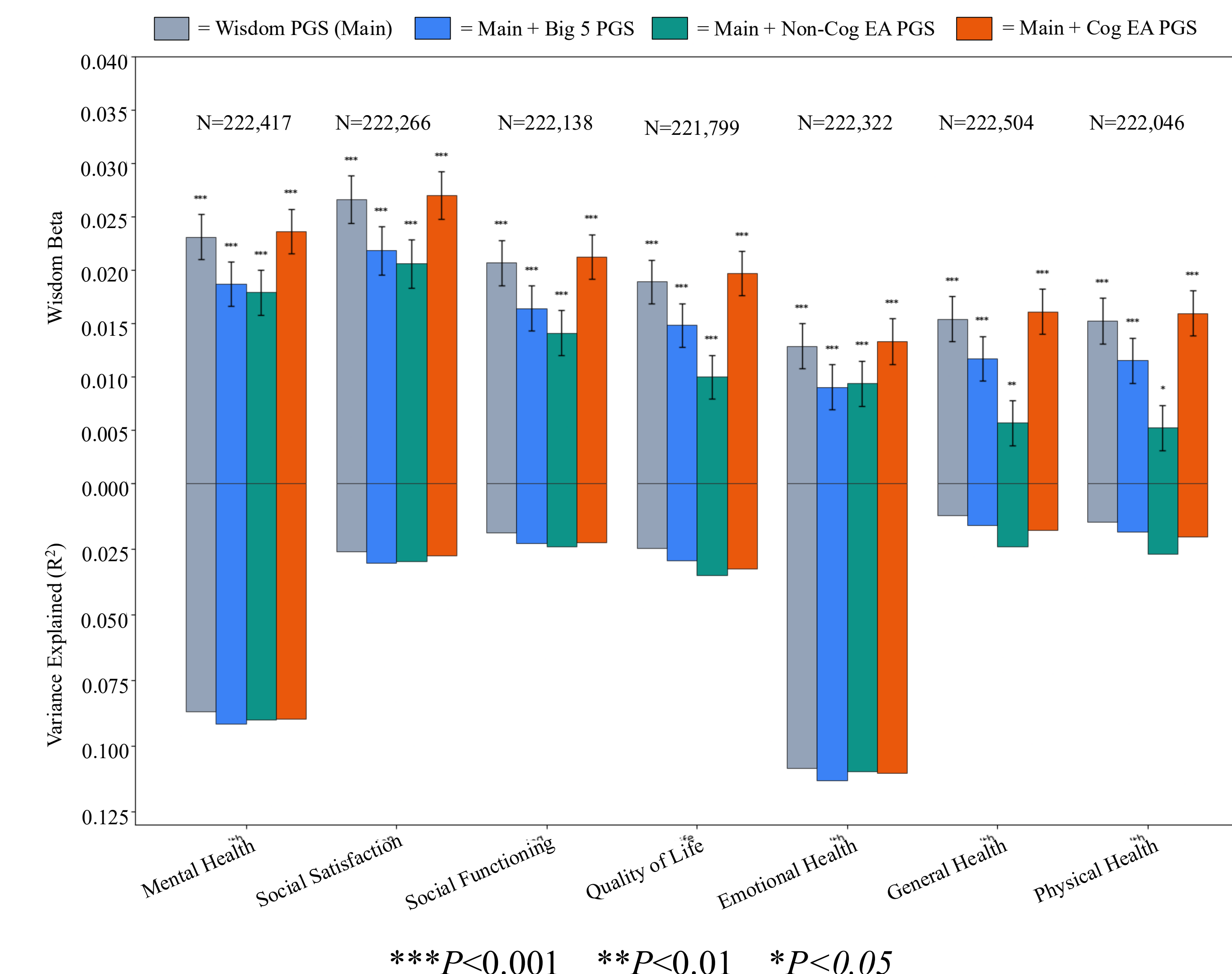


Figure 3: Predicting Psychosocial Outcomes. Polygenic scores derived from our GWAS are significantly associated with social and health outcomes in the independent *All of Us* cohort.

Summary and Future Directions

This study advances our understanding of wisdom as a biologically-grounded trait that could ultimately be used to develop informed interventions that increase wisdom and improve health and well-being.

Next Steps:

- Conduct item-level GWAS on the 7 individual SD-WISE components to map their independent genetic architectures.
- Increase discovery sample sizes to boost power and potentially map additional loci

Acknowledgments & References

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References

- [1] Thomas, M. L., et al. (2022). Abbreviated San Diego Wisdom Scale (SD-WISE-7) and Jeste-Thomas Wisdom Index. *Int Psychogeriatr*, 34(7).

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